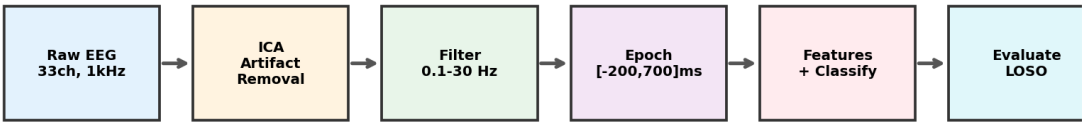
EEG-ERP Classification Pipeline

Oddball Paradigm: From Raw EEG to Target Detection

10	32	3,600	0.816
Participants	EEG Channels	Epochs	Best Accuracy

Pipeline Overview



Dataset: Oddball paradigm. Participants viewed a stream of stimuli with rare targets (1:5 ratio). EEG recorded at 1000 Hz from 32 channels. 60 target and 300 non-target epochs per participant.

Goal: Classify single EEG trials as target vs non-target using ERP signatures (P300, N170, VPP), comparing traditional feature-based methods against spatial filtering approaches.

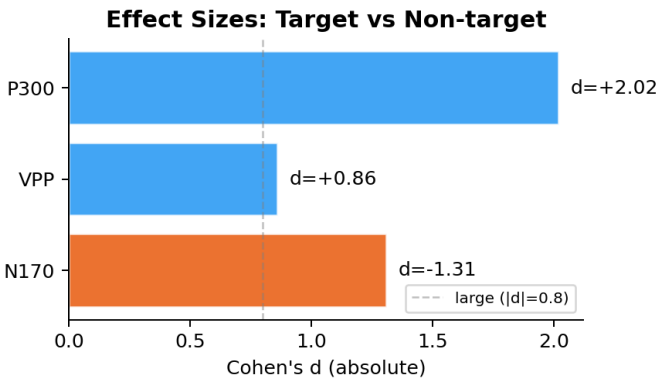
1. Preprocessing Pipeline

The pipeline uses a **two-track ICA strategy**: a heavily filtered copy (1 Hz high-pass, downsampled to 250 Hz) is used to train the ICA decomposition, while the original data receives gentle 0.1-30 Hz bandpass filtering. The ICA unmixing matrix transfers between the two because spatial channel relationships are filter-invariant. This gives clean ICA decomposition without distorting the slow P300 waveform.

Step	Operation	Why
1	Drop HEOG channel	Noisy hardware EOG - unreliable
2	ICA copy: HP 1 Hz, downsample 250 Hz	Slow drifts degrade ICA; speed up fitting
3	Fit ICA (31 components)	Rank = 32 channels - 1 (avg reference)
4	Surrogate EOG from Fp1/Fp2, F7/F8	Hardware HEOG dropped; use EEG proxies
5	Eye IC detection (2+ votes needed)	VEOG corr + HEOG corr + topography + ICLabel
6	Apply to 0.1-30 Hz filtered original	Remove artifacts, preserve ERP waveforms
7	Epoch [-200, 700] ms, baseline correct	900 ms window capturing N170 + P300

2. ERP Components & Statistical Results

Three ERP components were extracted from the cleaned epochs. All showed significant target vs non-target differences with large effect sizes:



Effect sizes for target vs non-target comparison (N=10, all p < 0.05)

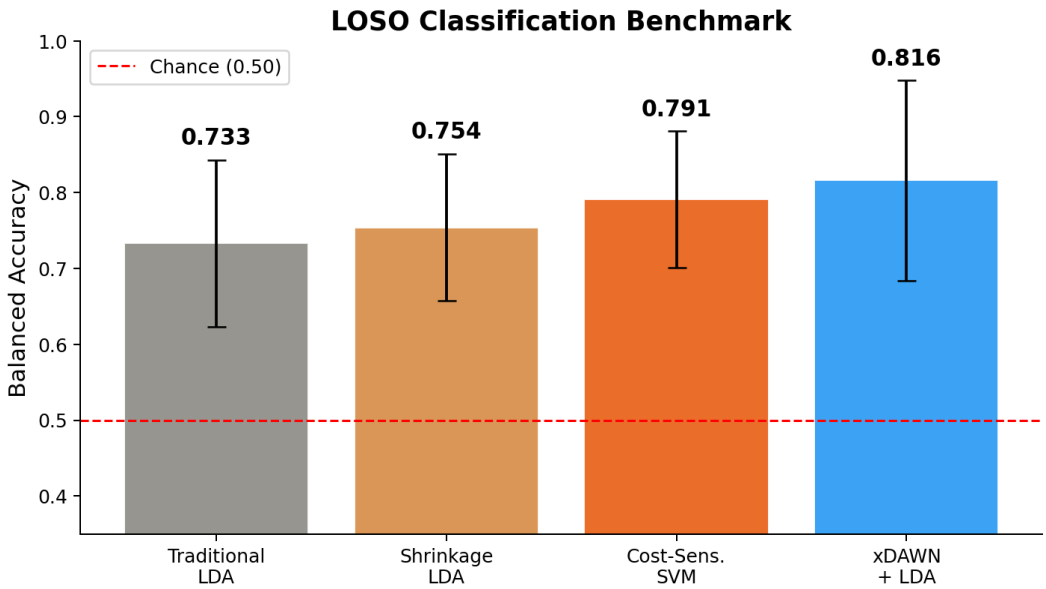
Component	Window	Channels	Target	Non-tgt	t(9)	p	Cohen's d
N170	130-200 ms	P7/P8/PO9/PO10	-2.13 uV	+0.59 uV	-4.13	.003**	-1.31
VPP	130-200 ms	Cz/Fz/FC1/FC2	+1.47 uV	-0.19 uV	2.72	.023*	+0.86
P300	250-500 ms	Pz/CP1/CP2/Cz	+4.30 uV	-0.07 uV	6.38	<.001***	+2.02

Wilcoxon signed-rank tests confirmed all results (P300: W=0, p=.002; N170: W=3, p=.010; VPP: W=6, p=.027)

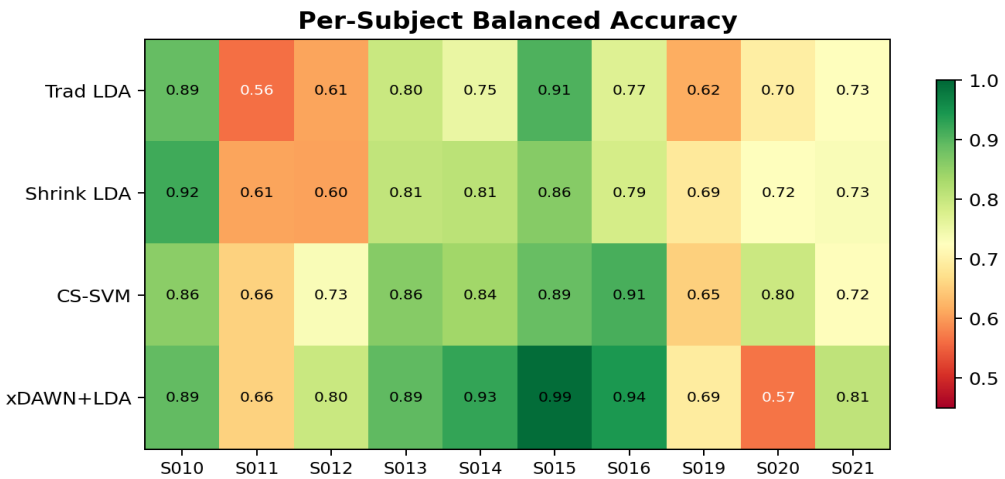
3. Classification: 4-Method Benchmark

Four classification approaches were compared using **Leave-One-Subject-Out (LOSO)** cross-validation. Each fold trains on 9 subjects and tests on the unseen 10th. This is the strictest test of cross-subject generalization.

#	Method	Input	Key Idea
1	Traditional LDA	64 features (mean amplitude)	Domain-knowledge baseline: mean ERP amplitude in N170 + P300 windows, all 32 channels
2	Shrinkage LDA	192 features (mean + peak + var)	Ledoit-Wolf regularization handles high dimensionality (192 features, 60 minority trials)
3	Cost-Sensitive SVM	192 features (mean + peak + var)	RBF kernel with class weights (5:1) for imbalance; C tuned via inner 3-fold GridSearch
4	xDAWN + Shrinkage LDA	Raw epochs (32ch x 701 times)	Supervised spatial filter maximizes ERP SNR; 4 filters compress 32ch to 8 virtual channels



LOSO balanced accuracy (mean +/- SD across 10 subjects). Red dashed line = chance level.



Per-subject breakdown. Green = high accuracy, red = near chance. Note sub-020 failure for xDAWN.

4. Key Insights

xDAWN wins (0.816) despite never being told where the P300 is. It *learns* the optimal spatial filter from data, compressing 32 channels into 8 virtual channels that maximize the target-evoked signal-to-noise ratio. This bridges the gap between data-driven spatial learning and regularized classification.

Domain knowledge still matters. Traditional LDA (0.733) uses hand-picked time windows and mean amplitude features from 60 years of ERP literature. It beats any method that ignores this knowledge. The best results come from combining learned spatial structure (xDAWN) with regularized classifiers (Shrinkage LDA).

Subject variability is the bottleneck. Sub-015 reaches 0.992 with xDAWN while sub-020 drops to 0.568. This 42-point gap on the same method reveals that inter-subject ERP topography differences, not the algorithm, limit cross-subject classification.

Rich features help. Shrinkage LDA with 192 features (mean + peak + variance) scores 0.754 vs Traditional LDA's 0.733 with 64 features. The additional peak amplitude and within-window variance features carry discriminative information beyond mean amplitude.

Wilcoxon confirms xDAWN's lead is statistically significant over Traditional LDA (p=.037) and all lower-ranked methods. The margin over Cost-Sensitive SVM (+0.025) is not significant (p=.121), suggesting these two approaches are comparably effective.

5. Practical Implications

P300 Brain-Computer Interfaces: xDAWN at 0.816 cross-subject balanced accuracy with only 60 target trials per user enables practical BCI spellers. With multiple flash repetitions per character, single-character accuracy would exceed 95%. The cross-subject success means reduced calibration time for new users.

Attention Monitoring: The 1:5 target-to-nontarget ratio mirrors real-world rare-event detection scenarios (air traffic control, surgical monitoring). The pipeline demonstrates feasible detection of whether an operator noticed a critical event.

Clinical Screening: P300 amplitude reduction is a biomarker for schizophrenia, Alzheimer's, and traumatic brain injury. The large effect size (d = 2.02) confirms robust detection with just 10 participants. The sub-020 failure pattern could itself be diagnostically informative - atypical ERP topography may indicate neurological differences.

Methodological Guidance: For small EEG datasets (N < 20), feature-based methods with domain knowledge outperform pure data-driven approaches. Use xDAWN when spatial filtering is appropriate; use Shrinkage LDA when computational cost matters (1.8s vs 169s). Reserve Riemannian methods for larger datasets.

Pipeline: MNE-Python + scikit-learn + pyriemann. 10 subjects, 32 EEG channels, oddball paradigm, 0.1-30 Hz, ICA artifact correction, [-200, 700] ms epochs, LOSO evaluation.